

3D Representations for World Modeling and Generation



Invited Speaker

Xiaojuan Qi

University of Hong Kong

Date: May 13, 2026 (Wed)

Time: 15:30 (HKT)

Venue: LAU-6-209, CityU

Biography

Xiaojuan Qi is currently an Associate Professor in the Department of Electrical and Electronic Engineering at the University of Hong Kong. She received her Ph.D. from the Chinese University of Hong Kong and has conducted research and academic exchanges at the University of Toronto, the University of Oxford, and Intel's Visual Computing Group. Her research focuses on empowering machines with the ability to perceive, understand, and reconstruct the visual world in open environments, with an emphasis on real-world deployment in embodied agents. She has authored over 100 papers in premier conferences across computer vision, graphics, and machine learning, including SIGGRAPH Asia, CVPR, ICCV, and NeurIPS, with many selected for oral presentations. Dr. Qi received a Best Paper Honorable Mention at SIGGRAPH Asia and was named one of IEEE AI's 10 to Watch in 2024, MIT TR 35 China. She actively serves the academic community, frequently as an Area Chair for ICCV, CVPR, NeurIPS, ICML, and AAAI.

Abstract

Humans perceive and interact with the world through rich 3D structure and dynamic visual experiences. Building AI systems with similar capabilities requires representations that jointly model geometry, appearance, and motion, while remaining efficient and editable. In this talk, I will present our recent progress on 3D/4D representations for world modeling and generation. I will first discuss advances in Gaussian splatting for real-time novel view synthesis and introduce structured extensions that enable high-quality surface reconstruction, motion modeling, and intuitive editing of dynamic scenes through sparse control and deformable kernels. I will then shift to world generation, presenting generative models for textured 3D assets and 3D-consistent video synthesis guided by explicit spatial representations. Together, these efforts move toward scalable, controllable, and physically grounded world models for spatial intelligence and embodied AI.